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date: 02-25-2019	HILIC Resuspension Solvent Preparation	Code no.: HILICResusSolv02252019

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HILIC Resuspension Solvent Preparation Table of Contents 1. Chemicals, Glassware, and Consumables 2. Equipment 3. Preparation of Stock Solutions 4. Preparation of HILIC Resuspension Solvent 5. Storage 6. Disposal of Waste 1. Chemicals, Glassware, and Consumables • D9-Caffeine, CDN Isotopes D-5972 • 12-[[[Cyclohexylcarbamoyl]amino]dodecanoic acid (CUDA), Cayman Chemical 10007923 • D4-Acetylcholine, CDN Isotopes D-2558 • D3-AC(18:0), Sigma Aldrich 793280 • D3-AC(16:0), Sigma Aldrich 730920 • D3-AC(14:0), Sigma Aldrich 730939 • D3-AC(8:0), Sigma Aldrich 730912 • D3-Creatinine, CDN Isotopes D-3689 • D9-Choline, shop.isotopes.com DLM-549-1 • D9-Trimethylamine Oxide (TMAO), shop.isotopes.com DLM-4779-1 • D3-1-Methylnicotinamide, TRC-Canada M323237 • Val-Tyr-Val, Sigma Aldrich 17355-22-5 • D9-Betaine, CDN Isotopes D-3352 • D3-AC(2:0), Sigma Aldrich 362049-62-5 • D3-Histamine, N-methyl-, CDN Isotopes D-2271 • D3-L-Carnitine, shop.isotopes.com DLM-1871-0.1 • D9-Butyrobetaine, CDN Isotopes, D-2271 • D9-Crotonobetaine, TRC-Canada C818802 • D3-Creatine, CDN Isotopes D-1972 • D3-DL-Alanine, CDN Isotopes D-1462 • D5-L-Glutamine, CDN Isotopes D-2532 • D3-DL-Glutamic Acid, CDN Isotopes D-1196 • D3-DL-Aspartic Acid, CDN Isotopes D-1953 • 15N2-L-Arginine, shop.isotopes.com NLM-395-0.5 • D4-Cystine, CDN Isotopes DLM-1000-PK • D4-L-Alanine, CDN Isotopes DLM-250-PK • D7-L-Arginine, CDN Isotopes DLM-541-PK • D3-L-Asparagine, CDN Isotopes DLM-6844-PK • D5-L-Glutamic Acid, CDN Isotopes DLM-556-PK • D5-L-Histidine, CDN Isotopes DLM-7855-PK • D10-L-Isoleucine, CDN Isotopes DLM-1505-PK • D10-L-Leucine, CDN Isotopes DLM-567-PK	

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- D8-L-Lysine, CDN Isotopes DLM-2641-PK
- D8-L-Methionine, CDN Isotopes DLM-6797-PK
- D2-L-Ornithine, CDN Isotopes DLM-4261-PK
- D8-L-Phenylalanine, CDN Isotopes DLM-372-PK
- D7-L-Proline, CDN Isotopes DLM-487-PK
- D3-L-Serine, CDN Isotopes DLM-582-PK
- D5-L-Threonine, CDN Isotopes DLM1693-PK (Change to 15N potentially)
- D8-L-Tryptophan, CDN Isotopes DLM-6903-PK
- D7-L-Tyrosine, CDN Isotopes DLM-589-PK
- D8-L-Valine, DLM-488-PK
- 42 x 20 mL Amber Glass Vials with screw lids, Gerstel P/O#: 093640-037-00
- 250 mL glass bottle with lid
- 100 mL graduated cylinder
- LC-MS grade Water, Fisher Chemical 4L W6-4
- LC-MS grade Acetonitrile (ACN), Fisher Chemical 4L A955-4
- LC-MS grade Methanol (MeOH), Fisher Chemical 4L A456-4

2. Equipment

- MiniVortexer (VWR)
- Centrifuge (Eppendorf 5415 D)
- VWR Sonication Bath: Model 75HT
- 10 mL pipette with appropriate pipette tips (S/N: 4345994)
- 100 µL pipette with appropriate pipette tips (S/N: L15354C)
- 200 µL pipette with appropriate pipette tips (S/N: M24713B)
- 1000 µL pipette with appropriate pipette tips (S/N: O19938B)

3. Preparation of Stock Solutions

3.1. Internal Standard Stock Solution Preparation

- a) Label each of the 42, 20 mL Amber Glass Vials with the full name of the internal standard, the concentration of the stock solution, the date the stock solution is prepared, and the solvent used to dissolve the internal standard. Names, concentrations, and appropriate solvents can be found in Table 1.
- b) Measure out the appropriate amount of raw internal standard into the appropriate amber vial. Record the actual amount of internal standard added to vial, as this number will be used to calculate the appropriate amount of solvent to add to the vial.
- c) Cap the vials while measuring other internal standards
- d) Acquire the following solutions, ensuring that any glassware used is clean and conditioned
 - 80 mL of (v/v) 80:20 ACN:Water
 - 15 mL of MeOH
 - 100 mL of (v/v) 50:50 ACN:Water
 - 250 mL of LC-MS grade Water
- e) The concentration of each stock solution needs to be maintained at the reported volume in Table 1. Therefore use the recorded weighed amount of each standard to calculate the exact volume of solution needed to maintain the concentration.
 - For example, if a stock solution of 0.1 mg/mL is required, 10 mL of solvent would ideally be added to 1 mg of standard. However, exactly 1 mg of standard is not

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realistic to measure. If 1.05 mg of standard is weight out, then 10.5 mL of solvent would need to be added to the glass vial instead of 10 mL.

- f) Following Table 1 add the appropriate amount of solvent to each of the internal standards. Ensure the correct solution is added to each internal standard. Volumes reported in Table 1 are ideal and need to be corrected to the exact amount of standard following step e).
- g) Vortex each internal standard lightly. Do not allow the solvent to touch the underside of the lid, as this may cause contamination.
- h) Sonicate each internal standard in room temperature water for 5 minutes.
- i) If solid has not dissolved, continue to repeat steps g) and h) until standard is fully dissolved.
- j) Store standards at 4°C

4. Preparation of HILIC Resuspension Solvent

4.1. Glassware Conditioning

- Condition a 250 mL glass bottle three times with 50:50 (v/v) ACN:Water.
- Condition a 100 mL graduated cylinder three times with 50:50 (v/v) ACN:Water
- If possible, dedicate glassware that is to only be used for HILIC Resuspension Solvent preparation.

4.2. HILIC Resuspension Solvent Preparation

- a) Add 63 mL of ACN to the 250 mL glass bottle.
- b) Add 14 mL of Water to the 250 mL glass bottle.
- c) Following Table 2, add the appropriate amount of each stock solution to the 250 mL glass bottle.
- d) Sonicate the Resuspension Solvent for 5 minutes in room temperature water.
- e) Do not allow the solution to touch the lid of the bottle, as this will cause plastic contamination.

5. Storage

5.1. Solid Standards

- Table 3 outlines the locations of the solid standards.
- Standards stored at room temperature should be kept sealed in Desiccator Box 1
- After all standards are opened, they should be tightly sealed with ParaFilm around the joint between the lid and the bottle.

5.2. Prepared Internal Standards

- All prepared internal standards should be stored at 4°C.
- Due to the high water content of many of the standards, do not freeze standards, as this will cause the water to freeze out of solution, potentially cracking the amber vial.

6. Disposal of Waste

- All waste should be disposed in non-halogenated waste containers that are clearly labeled with a waste label.
- Used pipette tips can be discarded in a beaker labeled "Used Pipette Tips". Once finished with entire SOP, beaker needs to be emptied into a larger plastic waste container with an appropriate waste label. Return the labeled beaker to a fume hood for later use.
- All used glassware should be rinsed three times with water, which is to be disposed in non-halogenated waste. After three washes, used glassware is placed on the dirty glassware cart.

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Table 1: Preparation of Stock Solutions

Standard	Solid (mg)	Solvent (mL)	Solvent	Concentration (mg/mL)
1_ISTD CUDA	1	10	MeOH	0.10
1_ISTD D3-Creatinine	1	10	H2O	0.10
1_ISTD D3-1-Methylnicotinamide	1	10	H2O	0.10
1_ISTD D3-Histamine, N-methyl-	1	10	H2O	0.10
1_ISTD D9-Butyrobetaine	10 mM	10	H2O	1.54
1_ISTD D9-Crotonobetaine	10 mM	10	H2O	1.52
1_ISTD D3-Creatine	1	10	H2O	0.10
1_ISTD D3-DL-Alanine	3	10	H2O	0.30
1_ISTD D3-DL-Glutamic acid	10	10	H2O	1.00
1_ISTD D3-DL-Aspartic acid	10	10	H2O	1.00
1_ISTD D4-Cystine	1.5	10	H2O	0.15
1_ISTD D3-Asparagine	5	15	H2O	0.33
1_ISTD D5-Histidine	1	5	H2O	0.20
1_ISTD D10-Isoleucine	1	10	H2O	0.10
1_ISTD D10-Leucine	2	15	H2O	0.13
1_ISTD D8-Lysine	2	15	H2O	0.13
1_ISTD D4-Alanine	3	10	H2O	0.30
1_ISTD D7-Arginine	10	10	H2O	1.00
1_ISTD D5-Glutamic acid	10	10	H2O	1.00
1_ISTD D5-Threonine	1	10	H2O	0.10
1_ISTD D8-Methionine	1	10	H2O	0.10
1_ISTD D2-Ornithine	4	15	H2O	0.27
1_ISTD D7-Proline	1	5	H2O	0.20
1_ISTD D3-Serine	2	15	H2O	0.13
1_ISTD D8-Tryptophan	1	10	H2O	0.10
1_ISTD D8-Valine	5	5	H2O	1.00
1_ISTD D9-Caffeine	1	10	ACN/H2O 80/20	0.10
1_ISTD D4-Acetylcholine	1	10	ACN/H2O 80/20	0.10
1_ISTD D9-Choline	1	10	ACN/H2O 80/20	0.10
1_ISTD D9-TMAO	1	10	ACN/H2O 80/20	0.10
1_ISTD D9-Betaine	1	10	ACN/H2O 80/20	0.10
1_ISTD D3-AC(2:0)	1	10	ACN/H2O 80/20	0.10
1_ISTD D3-L-Carnitine	1	10	ACN/H2O 80/20	0.10
1_ISTD D3-AC(18:0)	1	10	ACN/H2O 50/50	0.10
1_ISTD D3-AC(16:0)	1	10	ACN/H2O 50/50	0.10
1_ISTD D3-AC(14:0)	1	10	ACN/H2O 50/50	0.10
1_ISTD D3-AC(8:0)	1	10	ACN/H2O 50/50	0.10
1_ISTD Val-Tyr-Val	1	10	ACN/H2O 50/50	0.10
1_ISTD D5-L-Glutamine	10	10	ACN/H2O 50/50	1.00
1_ISTD 15N2-L-Arginine	10	10	ACN/H2O 50/50	1.00
1_ISTD D8-Phenylalanine	1	10	ACN/H2O 50/50	0.10
1_ISTD D7-Tyrosine	1	10	ACN/H2O 50/50	0.10

Note: For Butyrobetaine and Crotonobetaine, 10 mL of solvent is added to the shipped standard at 10 mM.

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Table 2: Volume of Stock Solution to add to Resuspension Solvent per 77 mL of 80:20 ACN:Water

Standard	Volume (μL)
1_ISTD CUDA	10
1_ISTD D3-AC(18:0)	10
1_ISTD D3-AC(16:0)	10
1_ISTD D3-AC(14:0)	10
1_ISTD D3-AC(8:0)	10
1_ISTD D9-Butyrobetaine	20
1_ISTD D9-Crotonobetaine	20
1_ISTD D5-Threonine	30
1_ISTD D3-AC(2:0)	40
1_ISTD D9-Caffeine	50
1_ISTD D4-Acetylcholine	50
1_ISTD D9-Choline	50
1_ISTD D9-TMAO	50
1_ISTD D3-Histamine, N-methyl-	50
1_ISTD 15N2-L-Arginine	50
1_ISTD D7-Arginine	50
1_ISTD D5-Glutamic acid	50
1_ISTD D4-Alanine	125
1_ISTD D3-Asparagine	125
1_ISTD Val-Tyr-Val	150
1_ISTD D3-Creatinine	200
1_ISTD D9-Betaine	200
1_ISTD D3-Creatine	200
1_ISTD D5-L-Glutamine	200
1_ISTD D3-1-Methylnicotinamide	250
1_ISTD D3-L-Carnitine	250
1_ISTD D3-DL-Glutamic acid	250
1_ISTD D2-Ornithine	300
1_ISTD D8-Methionine	400
1_ISTD D8-Tryptophan	450
1_ISTD D3-DL-Aspartic acid	500
1_ISTD D4-Cystine	500
1_ISTD D5-Histidine	500
1_ISTD D8-Phenylalanine	500
1_ISTD D10-Isoleucine	575
1_ISTD D7-Proline	600
1_ISTD D8-Valine	600
1_ISTD D3-DL-Alanine	750
1_ISTD D7-Tyrosine	750
1_ISTD D3-Serine	950
1_ISTD D8-Lysine	1000
1_ISTD D10-Leucine	1300

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Table 3: Solid Standards Storage Location and Conditions

Metabolite	Location
1_ISTD CUDA	-20°C
1_ISTD D3-AC(18:0)	-20°C
1_ISTD D3-AC(16:0)	-20°C
1_ISTD D3-AC(14:0)	-20°C
1_ISTD D3-AC(8:0)	-20°C
1_ISTD D3-AC(2:0)	-20°C
1_ISTD Val-Tyr-Val	Amino Acid Storage Box
1_ISTD D5-Threonine	Desiccator Box 1
1_ISTD D9-Caffeine	Desiccator Box 1
1_ISTD D4-Acetylcholine	Desiccator Box 1
1_ISTD D9-Choline	Desiccator Box 1
1_ISTD D9-TMAO	Desiccator Box 1
1_ISTD D3-Histamine, N-methyl-	Desiccator Box 1
1_ISTD 15N2-L-Arginine	Desiccator Box 1
1_ISTD D7-Arginine	Desiccator Box 1
1_ISTD D5-Glutamic acid	Desiccator Box 1
1_ISTD D4-Alanine	Desiccator Box 1
1_ISTD D3-Asparagine	Desiccator Box 1
1_ISTD D3-Creatinine	Desiccator Box 1
1_ISTD D9-Betaine	Desiccator Box 1
1_ISTD D3-Creatine	Desiccator Box 1
1_ISTD D5-L-Glutamine	Desiccator Box 1
1_ISTD D3-1-Methylnicotinamide	Desiccator Box 1
1_ISTD D3-L-Carnitine	Desiccator Box 1
1_ISTD D3-DL-Glutamic acid	Desiccator Box 1
1_ISTD D2-Ornithine	Desiccator Box 1
1_ISTD D8-Methionine	Desiccator Box 1
1_ISTD D8-Tryptophan	Desiccator Box 1
1_ISTD D3-DL-Aspartic acid	Desiccator Box 1
1_ISTD D4-Cystine	Desiccator Box 1
1_ISTD D5-Histidine	Desiccator Box 1
1_ISTD D8-Phenylalanine	Desiccator Box 1
1_ISTD D10-Isoleucine	Desiccator Box 1
1_ISTD D7-Proline	Desiccator Box 1
1_ISTD D8-Valine	Desiccator Box 1
1_ISTD D3-DL-Alanine	Desiccator Box 1
1_ISTD D7-Tyrosine	Desiccator Box 1
1_ISTD D3-Serine	Desiccator Box 1
1_ISTD D8-Lysine	Desiccator Box 1
1_ISTD D10-Leucine	Desiccator Box 1
1_ISTD D9-Butyrobetaine	Order Fresh
1_ISTD D9-Crotonobetaine	Order Fresh